클래스 다이어그램 — backend-api (Spring Boot 3.3.5 · MyBatis)

com.dronecontrol.<도메인> 패키지별 Controller → Service → Mapper(@Mapper SQL 어노테이션) 구조와 서비스 간 의존 (주요 메서드만 표기)

UserController · AuthController

+register() POST /users
+login() / logout()
+me() · list(role) · updateProfile()
+approve() · reject()
+changePassword() · changeRole()
+delete()
-isAdmin() · isSuperAdmin()

ZoneController · RegionController

+zones CRUD · neighbors(PUT/GET)
+getStats() · exportStats()
+streamStats(): SseEmitter
+getPlayback() · thresholds(GET/PUT)
+regions CRUD · flows()

CameraController

+list(zoneId) · get() · create()
+update(): 가동 중 소스 변경 409
+changeState(): 400/404/502

InternalStatsController
InternalCameraStatusController

+receive(List<StatSample>) → 204/404
+reportStatus() → 204/404
(/internal/v1, 인증 없음)

EventController · WorkHistoryController

+list(zoneId, status) · get()
+updateStatus(): 로그인만 확인(401)
+history(): SUPER_ADMIN 외 403

checkAndRecord

UserService

+register(): 최초 SUPER_ADMIN
+login(): PasswordUtil.matches
+approve()→USER · reject()→PRE_USER
+changeAdminRole()
+changePassword(requireCurrent)

ZoneService · RegionService

+setNeighbor(): 반대 방향 자동 등록
(N↔S, E↔W, 자기 자신 거부)
+create/update/list

CameraService

+changeState(RUNNING|STOPPED)
RUNNING → startAnalysis → PREPARING
DUMMY면 recordPlaybackStart()
+reportStatus(status, progress)

StatsService

+recordSample(): density 계산·UPSERT
→ eventService.checkAndRecord()
→ zoneFlowService.recordFlows()
→ broadcast SSE 'stat'
+subscribe(zoneId): SseEmitter(10분)
+getMinuteStats()

EventService

+checkAndRecord(zoneId, camId, density)
가장 높은 넘은 단계 · 미처리 있으면 생략
+getThresholds(): 기본 0.01/0.02
+setThresholds(): 삭제 후 재삽입

UserMapper «@Mapper»

insertUser · selectByLoginId
selectById · selectByRole
countSuperAdmins · updateRole
updatePassword · deleteUser

ZoneFlowService

+recordFlows(zoneId, at, stat)
outflow_N/E/S/W → 인접 구역
+getFlowsByDong()

RecordingService · RecordingMapper

+recordPlaybackStart(cameraId, uri)
+getRecordings(zoneId, from, to)

StatsExportService

+buildWorkbook(): Apache POI
15개 열 · 헤더 스타일 · 틀 고정

WorkHistoryService

+log(adminId, actionType, target, detail)
+list(adminId, actionType)

AnalysisServerClient

+startAnalysis(): POST → 202 확인
+stopAnalysis(): DELETE
base-url http://localhost:5050

PasswordUtil

+hash(): SHA-256 hex
+matches(plain, hashed)

ZoneMapper · ZoneFlowMapper
RegionMapper «@Mapper»

insertZone · selectZonesByDongId
upsertNeighbor · selectNeighbors
upsertFlow · selectFlowsByDong

StatsMapper «@Mapper»

selectZoneAreaByCameraId
upsertMinuteStat (ON DUPLICATE KEY)
selectMinuteStats · selectMinuteStatAt

EventMapper · WorkHistoryMapper

insertEvent · selectOpenEventForZone
updateStatus · thresholds upsert/delete
work_history insert/select

User

id · loginId · @JsonIgnore passwordEncoded
email · phone · name
role(PRE_USER/USER/ADMIN/SUPER_ADMIN)

Zone · ZoneNeighbor · ZoneFlow · Region

Zone(id, dongId, name, areaM2, geometry)
ZoneNeighbor(direction, neighborId)
Region(id, parentId, level, code, name)

Camera · Recording

Camera(id, zoneId, name, sourceType,
sourceUri, status, progress, errorMessage)
Recording(id, cameraId, startedAt, filePath)

StatSample · ZoneMinuteStat

StatSample(cameraId, at, personCount,
in/out{N,E,S,W}, dwellSec[], avgSpeedPxS, moves)
ZoneMinuteStat(count·density·flow·move 컬럼)

Event · EventThreshold · WorkHistory

Event(zoneId, cameraId, levelNo, density,
threshold, status, handledBy, memo)
EventThreshold(zoneId, levelNo, densityThreshold)

남색 = Controller(@RestController) · 파랑 = Service(@Service) · 주황 = 외부 연동(@Component) · 초록 = Mapper(@Mapper) · 회색 = 도메인 POJO · 점선 = 의존(호출) · 주황 곡선 = CameraService → AnalysisServerClient · 컨트롤러는 WorkHistoryService.log()로 작업 이력 기록